Spatial-Spectral Contrastive Self-Supervised Learning With Dual Path Networks for Hyperspectral Target Detection

Xi Chen, Yuxiang Zhang, *Senior Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Deep learning (DL) has shown great success in hyperspectral target detection (HTD). Because of the limited targets of interest in the hyperspectral image (HSI), many DL-based methods focus on spectral feature extraction by expanding samples, exhibiting low model transferability, as well as high time consumption. To address these issues, we propose a novel spatial-spectral contrastive self-supervised learning-based HTD method with dual path networks (DPN-CSSTD). This method constructs a contrastive task using an HSI augmentation method and a DPN encoder on a large amount of unlabeled data, enabling the pretrained model with spatial-spectral discrimination ability. The DPN encoder uses the convolution neural network and skip connections to improve the spatial-spectral feature extraction. Additionally, the proposed method constructs a transfer learning model with the pretrained DPN encoder and a newly added detector to improve the pretrained model transferability and the time efficiency of target detection (TD) tasks. The transfer learning model is fine-tuned via a proposed fine-tuning strategy with a few selected samples to adapt to different TD tasks. Extensive experiments conducted on four challenging hyperspectral datasets demonstrate that the proposed method outperforms other state-of-the-art approaches.**

***Index Terms*— Contrastive learning, convolution neural network (CNN), hyperspectral target detection (HTD), self-supervised learning (SSL), simple linear iterative clustering (SLIC) segmentation.**

I. Introduction

A HYPERSPECTRAL image (HSI) is a 3-D cube that provides rich spectral information and spatial information [1]. HSI usually contains hundreds or even thousands of spectral bands with high spectral resolution. Hence, HSI is conducive to distinguishing subtle differences between different ground targets, providing a unique advantage to target detection (TD) tasks. Hyperspectral TD (HTD) aims to separate the target of interest from the dominant background with few priori target spectra [2]. Essentially, HTD can be regarded as a binary classification task, with the difference being that the number of targets is typically small and is much smaller than that of backgrounds. Therefore, HTD is considered to be a fundamental task and has sparked a great wave of applications in various scenarios, including disaster monitoring, medicine, agriculture, and military fields [3], [4], [5], [6], [7].

In the past few decades, many HTD methods have been proposed based on the spectra difference between target and background. Traditional detectors are constructed mainly based on spectral matched models, statistical distribution hypothesis models, and machine learning models [8]. The spectral matched model employs a matched filter or projection operator to highlight the target while suppressing the background, such as constrained energy minimization (CEM) [9], orthogonal subspace projection (OSP) [10], hierarchical CEM [11], and oblique projection [12]. The statistical distribution hypothesis model usually introduces a certain statistical distribution assumption for the target and the background to construct a generalized likelihood ratio-based detector, including an adaptive coherence estimator (ACE) [13], adaptive matched filter (AMF) [14], and matched subspace detector (MSD) [15]. Besides, many machine learning models are introduced to implement TD tasks, including kernel techniques [16], [17], metric learning [18], [19], sparse representation [20], [21], [22], tree-structured encoding [23], and so on [24]. These classical detectors are usually simple and mainly focus on extracting shallow spectral features for HTD, which can detect most targets in HSI.

Since the beginning of the deep learning (DL) era, many DL models have been introduced to learn rich and deep features in HSIs, such as convolution neural networks (CNNs) [25] and transformers [26], [27]; however, the performance of DL models largely depends on the size and quality of the training data. Lack of labeled training samples is a major challenge for TD tasks since there are usually only one or a few priori target spectra [28]. This limitation greatly impedes the applicability of DL in TD tasks. Many methods have been proposed to improve the training efficiency of DL models that lack of sufficient labeled training samples. The typical way is to employ various sample expansion strategies to generate pseudo labeled samples. For example, the typical TD model using deep network (HTD-Net) [29] uses a pixel-pairs strategy by applying the modified autoencoder (AE) to generate target

Manuscript received 4 January 2024; revised 19 March 2024; accepted 15 April 2024. Date of publication 18 April 2024; date of current version 30 April 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62071438, Grant 62222116, Grant 62225113, Grant 61701452, and Grant 62171417. *(Corresponding author: Yuxiang Zhang.)*

Xi Chen and Yuxiang Zhang are with the School of Geophysics and Geomatics, China University of Geosciences, Wuhan 430074, China (e-mail: chenxi29@cug.edu.cn; zhangyx@cug.edu.cn).

Yanni Dong is with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: dongyanni@whu.edu.cn).

Bo Du is with the School of Computer, Wuhan University, Wuhan 430079, China (e-mail: gunspace@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3390946

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

samples and the linear prediction strategy to select background samples. Literature [30] proposes the two-stream convolutional network (TSCNTD), which introduces the sparse representation to select background samples and mixes them with priori target spectra to generate mixed target samples. Literature [31] proposes a Siamese transformer network (STTD), which uses endmember extraction, unmixing, and refiltering to obtain the background samples and uses mixing operations to generate target samples. Literature [32] proposes a DL-based unconstrained linear mixture model (ULMMDL), which uses spatial segmentation and spectral clustering operation to obtain the pseudo labeled training samples. Literature [33] proposes a DL-based band selection method (DLBSTD), which develops a band selection-based correlation measurement strategy to select the background samples and a target band random substitution strategy to augment the target samples. Literature [34] proposes a weakly supervised HTD framework based on a transformer (HTDFormer), which obtains the training samples through a classic detector. Literature [35] and literature [36] use generative models adversarial AE (AAE) and auxiliary generative adversarial network (GAN) to generate the pseudo labeled training samples. In addition, self-supervised learning (SSL) has garnered attention for extracting information directly from the data itself without labeled samples. Literature [37] proposes a self-supervised spectral-level contrastive learning method (SCLHTD), which trains the model without using labeled samples to gain the capability of discriminating spectral differences. Literature [38] proposes a deep clustering-based self-supervised method (SSDCTD), which divides subsets through classic detectors and mines the discriminative features of targets and backgrounds by spectral deep clustering.

The above-mentioned DL-based methods have successfully broken through the barrier of limited labeled samples between DL architectures and HTD. These methods are, however, trained and tested on the same dataset, which can learn some knowledge of a specific TD task and cannot be effectively transferred to various TD tasks. This may lead to high time consumption, especially when faced with multiple TD datasets. Some researchers explore the transferability of models. Literature [39] uses a CNN architecture trained on labeled pixel pairs from hyperspectral classification datasets for HTD. Literature [40] proposes a semi-supervised domain adaptive few-shot learning method (SDAFL), which trains on abundant labeled data from the hyperspectral classification dataset and transfers the trained model to target domain. Literature [41] proposes meta-learning with the Siamese network method (MLSN), which also pretrains on abundant labeled data from the hyperspectral classification dataset and then transfers the pretrained model to TD tasks via fine-tuning the detection model; however, they still require a large number of labeled samples for pretraining, and the last two methods require additional filter postprocessing to obtain the final results. In addition, most of these DL-based methods are focused on spectral information extraction, paying less attention to spatial information extraction.

Based on the above analysis, there is a strong need to design a DL-based HTD method that can significantly reduce the demand of labeled or pseudo labeled samples, effectively extract the spatial-spectral features, and can be flexibly transferred to different HTD datasets. Inspired by contrastive self-supervised learning (CSSL) [42], we propose a spatial-spectral CSSL-based HTD method with dual path networks (DPNs), which can be labeled as DPN-CSSTD. This method constructs a contrastive task with a new HSI augmentation method and a DPN encoder on a large amount of unlabeled data, enabling the pretrained model with spatial-spectral discrimination ability, and the proposed method constructs a transfer learning model to effectively transfer the pretrained model to various downstream TD tasks with limited pseudo labeled samples.

The overall architecture of DPN-CSSTD includes two parts: spatial-spectral contrastive self-supervised pretraining and transferring the model to TD tasks. First, the contrastive task is performed via an HSI augmentation method and a DPN encoder on the positive samples and negative samples to obtain discrimination capabilities of spatial-spectral similarity and dissimilarity. The HSI augmentation method is designed to generate semantically identical patches from unlabeled HSI and combine the patches with their central spectra to form positive samples and negative samples. The DPN encoder is designed based on CNN and skip connections to learn spatial and spectral features of HSIs. Finally, we construct a transfer learning model with DPN and a newly added detector, which is fine-tuned using a strategy with a few selected pseudo labeled samples via the CEM detection method and simple linear iterative clustering (SLIC) [43] segmentation method; thus, it can effectively transfer the pretrained model to various TD tasks.

Compared with the strategies of generating numerous pseudo labeled samples by AE, linear mixing models or other methods, the proposed sample selection strategy can significantly reduce the demand for pseudo labeled training samples while maintaining the sample purity. Compared with the previous feature extraction model with sufficient training samples, the DPN encoder in the CSSL pretraining model can extract spatial-spectral features without labeled samples. Compared with the above DL-based single-scene HTD methods, the proposed DPN-CSSTD method can pretrain on one image scene and test on other various scenarios, which is beneficial for improving the model transferability and reducing the time computation, and compared with the above cross-scenario methods, the proposed DPN-CSSTD method does not rely on resource labeled samples and filter postprocessing operations.

The main contributions of our research are summarized as follows.

1) The novel dual path networks DPN are proposed via CNN and skip connections, which can effectively learn spatial and spectral features of HSI.
2) A contrastive task is constructed with a new HSI augmentation method, where the self-supervised signals are constructed by designing the contrastive task to train the pretraining model with spatial-spectral discrimination ability.
3) A transfer learning model with DPN and a detector is constructed, which is fine-tuned using a strategy with few selected pseudo labeled samples and can effectively transfer the pretrained model to various HTD tasks.

II. Related Work

A. Contrastive Self-Supervised Learning

SSL mainly uses pretext tasks to mine its own supervision information from large-scale unlabeled data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

This greatly reduces human annotation and produces numerous self-supervised data for training, resulting in the improvement in the learning of feature representation. As a branch of SSL, contrastive SSL (CSSL) [42] has achieved unprecedented performance in computer vision (CV) [44], [45], [46], natural language processing (NLP) [47], [48], and other fields. CSSL learns the invariance of different augmented data. It generates and compares positive and negative samples from augmented data [49], learning more discriminative feature representation by making positive samples close to each other and negative samples away from each other [50]. The learned feature representation is valuable for various downstream tasks. The specific process of CSSL is to first perform data augmentation on the image and construct positive and negative samples between the augmented views. Then, the feature representation is obtained through the encoder network and constrained by the contrastive loss function. Finally, the pretrained network is transferred to downstream tasks.

In recent years, some CSSL methods have been proposed in CV, such as momentum contrast (MoCo) for unsupervised visual representation learning [44], bootstrap your own latent (BYOL) [45], and similar contrastive learning of visual representations (SimCLR) [46]. MoCo uses momentum encoders to construct a dynamic dictionary by encoding positive and negative sample features as keys. The momentum encoder is updated based on the query encoder. The encoders for query and key are different and the goal is to make the query as similar as possible to the features of the key it matches. BYOL directly removes negative samples on the basis of MoCo and adds projections to two different networks. After updating the online network, a predictor is added to predict the results obtained by the target network, and the mean-squared-error (MSE) loss is used for constraints during network training. SimCLR uses ResNet as the backbone of the encoder and adds a projection head after the features obtained by the encoder [46]. It uses the normalized temperature-scaled cross entropy (NT-Xent) for constraints.

The pretraining models via CSSL can be transferred to various HSI fields, including image classification, change detection, TD, and so on [37], [51], [52], [53], [54], [55]. We follow the SimCLR framework and propose a spatial-spectral contrastive self-supervised pretraining model with a new HSI augmentation method and a DPN encoder, which constructs self-supervised signals by designing the contrastive task to train the pretraining model with spatial-spectral discrimination ability.

B. Convolutional Neural Network

Convolutional neural networks (CNN) are inspired by biological visual cognitive mechanisms, which can extract local features of objects by convolutional operations. A typical CNN usually includes the convolution layer, the pooling layer, the full connected (FC) layer, and the activation function [56], [57], [58]. The convolutional layer consists of several convolutional kernels that perform the convolution operation on input data. The shallow convolutional layer is responsible for extracting low-level features, such as edges, lines, and textures. The deep convolutional layer can iteratively extract top-level semantic features from low-level features multiple times. The pooling layer is to cut the features into several regions and take their maximum or average values to obtain new features with smaller sizes, thereby reducing the number of parameters. The FC layer combines all local features into global features to calculate the final score for each class. It is often used as the output layer of the network in TD tasks. The activation function is to increase the nonlinearity of the network. It is usually located behind each convolutional layer and used as the final detector for TD tasks. Based on CNN, we introduce skip connections to design the dual path networks with a spatial path and a spectral path to improve the extraction of local spatial features and spectral features.

C. Superpixel Segmentation

Superpixel refers to an irregular block of pixels composed of adjacent pixels with similar texture, color, brightness, and other features, which has certain visual significance [59]. SLIC [43] is a simple and easy to implement method. The process is mainly divided into four steps. Firstly, select the superpixel centers evenly within the image according to the number of superpixels. Move the superpixel center to the location corresponding to the minimum gradient position in a 3×3 neighborhood. Then, assign each pixel to the nearest superpixel center in a search region. After that, update the superpixel center to be the mean vector of all the pixels belonging to the superpixel. Repeat the assignment and update steps until the superpixel centers no longer change. By default, only the number of superpixels needs to be adjusted. In addition, SLIC is the most ideal in terms of running speed, compactness in generating superpixels, and contour preservation compared to other superpixel segmentation methods [43]. We, therefore, use SLIC to select representative background samples from the entire image.

III. Method

Our method can be divided into two parts, spatial-spectral contrastive self-supervised pretraining and transferring the pretrained model to TD tasks. The overall architecture of DPN-CSSTD is shown in Fig. 1.

A. Spatial-Spectral Contrastive Self-Supervised Pretraining

The pretraining model is trained on a large amount of unlabeled data to learn discrimination ability through a contrastive task. It is divided into four parts, namely HSI augmentation, dual path networks encoder, projection head, and contrastive loss. First, we use an HSI augmentation method to generate positive samples and negative samples, which serve as the input of the network. These samples are then fed into the designed DPN encoder to learn feature representations. Subsequently, the encoded features are projected onto a lower-dimensional representation space through a projection head. The contrastive loss is applied to constrain the model training.

1) Hyperspectral Image Augmentation: The first key step in contrastive self-supervised pretraining is to obtain positive and negative samples by image augmentation on a general HSI. In order to use the spatial and spectral information of all pixels, we perform edge filling on the original image and divide the image into regular patches centered on each pixel. The size of the patch is $w \times w$, which is experimentally set as 5 in this work. We then perform the data augmentation on these patches to generate positive and negative samples.

How data augmentation is conducted deserves further consideration since there is a lack of systematic image augmentation research in the HTD field. Based on the research

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

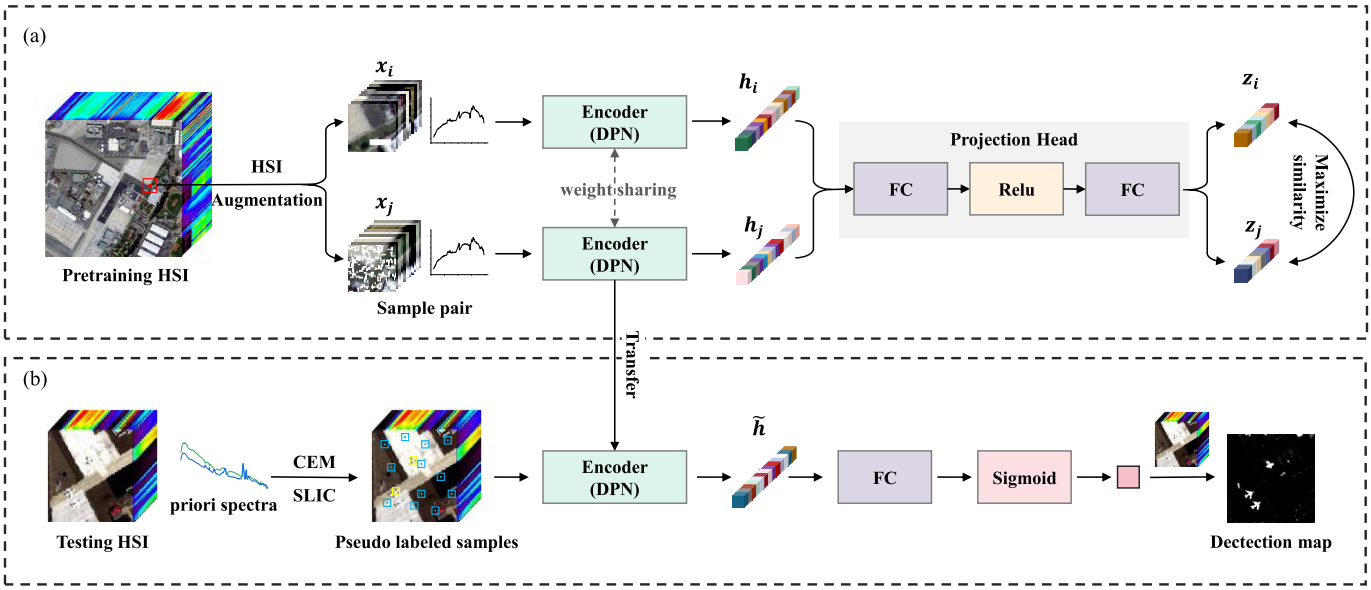


Fig. 1. Overall architecture of the proposed DPN-CSSTD method. (a) Spatial-spectral contrastive self-supervised pretraining. (b) Transferring to TD.

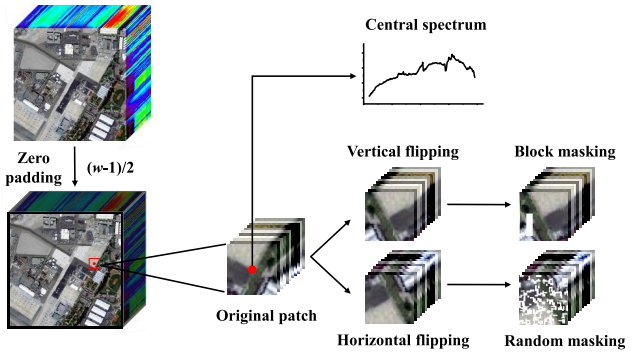


Fig. 2. Hyperspectral image augmentation in the pretraining stage.

of SimCLR, it can be found that the composition of data augmentation plays a key role in defining effective research of SimCLR, it can be found that the composition of data augmentation plays a key role in defining effective prediction tasks, and contrastive learning needs stronger data augmentation than supervised learning. What is more, the combination of different data augmentation methods can increase the difficulty of prediction but can also significantly improve the quality of representation.

We, therefore, design a simple spatial augmentation method, as shown in Fig. 2. It is a combination of two data augmentation strategies, including flipping (vertical flipping and horizontal flipping) and masking (block masking and random masking). A patch from the original image via different data augmentation strategies will produce semantically identical patches, forming positive sample pairs, and two augmented patches originated from different patches will generate semantically different patches, forming negative sample pairs. These positive and negative samples can be used to pretrain a specific model, as do in traditional CSSL methods; however, this way focuses on general spatial representation with spatial patches. In order to explore the abundant spectral information in HSI, the spectrum of the central pixel in the patch is also remained for representation learning. In summary, the sample pair labeled as x_i and x_j consists of two augmented patches and the central spectrum of the original patch.

2) Dual Path Networks Encoder: The designed DPN is used as the encoder for feature extraction. The architecture of the networks is shown in Fig. 3. The networks include a spectral path and a spatial path. The two paths aim to effectively extract spectral and spatial features by CNN and skip connections, where the spectral path pays attention to the continuity between the spectral dimensions, while the spatial path fully explores the spatial correlation of the patch.

The spectral path consists of six convolution layers, and each convolution layer is followed by the ReLU activation function. This path first performs dimension reduction on the spectrum to remove redundant information. This step uses a convolution operation with a kernel size of 1×7 and a step size of 2. In order to concatenate with the output spatial features of the spatial path, a convolution operation with a kernel size of $1 \times$ (band-6)/2 is added at the end of the spectral path to transform the output of the spectral path to a feature vector.

The spatial path consists of five convolution layers and a global average pooling (GAP) layer. Each convolution layer is followed by the ReLU activation function. This path first performs convolution operation on the spectral dimension with the input patch of convolution kernel of $1 \times$ band size. This operation reduces the spectral dimension of the input patch to 1, achieving the goal of only focusing on learning spatial features. Similarly, to concatenate with the output spectral features of the spectral path, a GAP operation is performed after the convolutional layers of the spatial path to obtain a spatial feature vector.

Both paths introduce skip connections in the first two to five convolutional layers and use concatenate operations. The difference is that the spectral path uses long-skip connection to concatenate the features of all previous layers and input them into the next layer, as shown in 1D-block in Fig. 3. The spatial path applies short-skip connection, which concatenates the input and output of this layer and inputs it to the next layer, as shown in 2-D block in Fig. 3. The skip connection operation retains more positional and dimensional information, enabling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

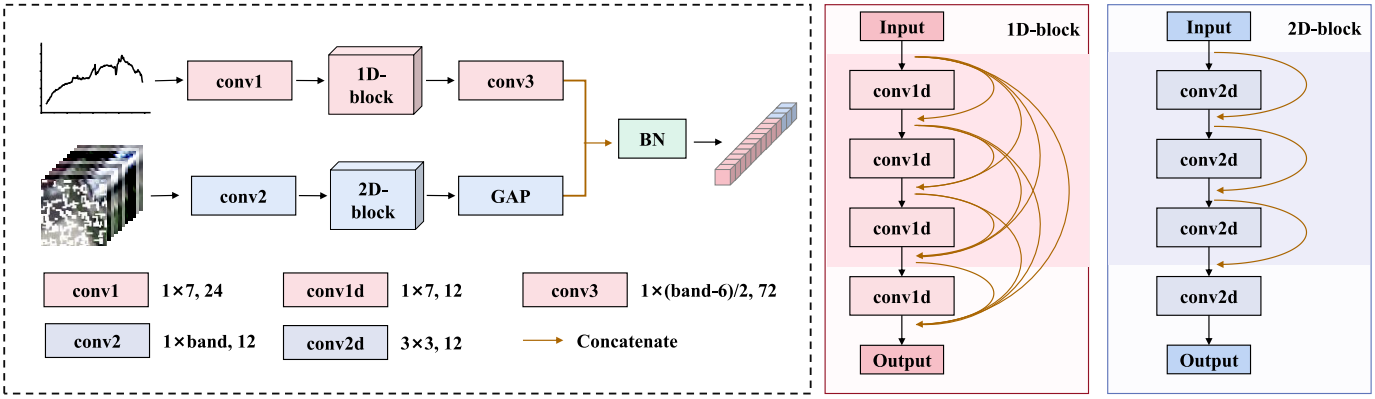


Fig. 3. Architecture of the dual path networks.

the next convolutional layer to freely choose between shallow features and deep features, strengthening the transmission of features [60], improving the reuse rate of feature information.

The output of the mth layer in the spectral path and the nth layer in the spatial path can be represented as:

$$y_m = H_m\left(\left[y_1, \ldots, y_{m-1}\right]\right) \quad m \in \{3, 4, 5, 6\} \tag{1}$$

$$y_n = H_n\left(\left[y_{n-2}, y_{n-1}\right]\right) \quad n \in \{3, 4, 5\} \tag{2}$$

where H_m and H_n are the nonlinear mapping of the mth and nth layer in two paths. $[y_1, \ldots, y_{m-1}]$ is the merging of the output features of the first $m-1$ layers in the spectral path and $[y_{n-2}, y_{n-1}]$ is the merging of the output features of layers $n-2$ to $n-1$ in the spatial path.

Finally, the features obtained from the two paths are concatenated and then normalized using the batch normalization (BN) layer, as the data distribution in each layer is different. This can make it easier for the training to converge. The finally encoded features for the sample pair x_i and x_j are labeled as h_i and h_j with a dimension of 84, where the dual path network outputs the features with a dimension of 72 and 12, respectively, from the spectral path and the spatial path.

3) Projection Head: The projection head includes two FC layers and a ReLU activation function. The projection head further maps the encoded features h_i and h_j of 84 dimensions to a low-dimension representation space to obtain new features z_i and z_j. The new feature dimensions are experimentally set as 32. The three-layer nonlinearity network can not only improve the representation quality of the DPN encoder but also avoid losing some important features due to the following contrastive loss calculation during training.

4) Contrastive Loss: The NT-Xent loss is used to constrain the network training, which aims to draw the positive samples closer together and the negative samples farther apart [46]. The NT-Xent loss for a batch sample pair can be calculated as follows:

$$\text{sim}\left(z_i, z_j\right) = \frac{z_i \cdot z_j}{||z_i||\,||z_j||} \tag{3}$$

$$L(i, j) = -\log \frac{\exp\left(\text{sim}\left(z_i, z_j\right)/\tau\right)}{\sum_{k=1, k\neq i}^{2N} \exp(\text{sim}(z_i, z_k)/\tau)} \tag{4}$$

where $\text{sim}(\cdot)$ is the cosine similarity, τ is the temperature coefficient and N is the batch size. The final loss obtained by calculating the average loss of all pairs in each batch is as follows:

$$L_{\text{NT-Xent}} = \frac{1}{2N} \sum_{k=1}^{N} [L(2k-1, 2k) + L(2k, 2k-1)]. \tag{5}$$

B. Transfer Learning

The transfer learning task is to transfer the pretrained model to TD tasks. The transfer learning model includes three parts, namely pseudo labeled sample selection, detection network with fine-tuning, and BCE loss. First, a few samples are selected based on CEM and SLIC segmentation. Then, the detection network with the pretrained DNP encoder and a detector is fine-tuned to adapt to various TD datasets. The BCE loss is applied to constrain the fine-tuning process. Once the transfer learning network is fine-tuned, each tested pixel and its image patch in a specific TD dataset are fed into the fine-tuned network, and the output values of the whole network are obtained as the detection statistics.

1) Pseudo Labeled Sample Selection: A few target samples are selected based on the initial detection results by CEM and a few background samples are selected via the SLIC segmentation method.

CEM is a classic HTD method and was chosen as the initial detector. The results obtained via CEM are arranged in descending order. Then, the first N_t pixels and their corresponding patches are selected to generate the target samples. The labels for these samples are set as 1. N_t is analyzed in Section IV. The CEM detector is denoted as follows:

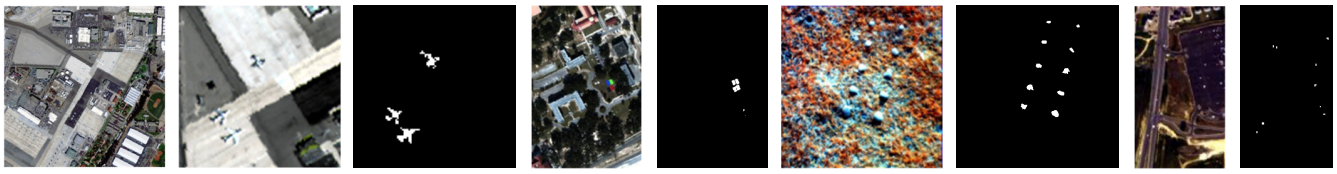
$$w^* = \frac{R^{-1}d}{d^T R^{-1} d} \tag{6}$$

$$D_{\text{CEM}}(x_s) = x_s^T w^* = \frac{x_s^T R^{-1} d}{d^T R^{-1} d} \tag{7}$$

where w^* is the filter, R is the autocorrelation matrix, d is the prior spectrum of the target, and x_s is the spectrum of the tested pixel.

In order to guarantee the representativeness of the background samples, we select the centroid pixels and their corresponding patches from superpixels via SLIC segmentation, forming the background samples. The labels for these samples are set as 0. The SLIC segmentation threshold is set according to the number of background samples, which is analyzed in Section IV. Considering the possibility of selecting the target pixels, we remove the centroid pixel that is spectrally

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Experiment datasets. (a) San Diego dataset for pretraining. (b) Image scene of AVIRIS dataset. (c) Ground-truth of AVIRIS dataset. (d) Image scene of Muufl dataset. (e) Ground-truth of Muufl dataset. (f) Image scene of Nuance Cri dataset. (g) Ground-truth of Nuance Cri dataset. (h) Image scene of HYDICE dataset. (i) Ground-truth of HYDICE dataset.

close to the prior target spectrum. The similarity is measured via the spectral angle distance:

$$\angle(d, x_c) = \arccos \frac{d^T x_c}{||d||\ ||x_c||} \tag{8}$$

where d is the prior spectrum of the target and x_c is the spectrum of each centroid pixel.

2) Detection Network With Fine-Tuning: The detection network consists of the pretrained DNP encoder and a detector with a FC layer and sigmoid activation function. In the fine-tuning process, the encoder is responsible for learning the features of the pseudo labeled samples. The FC layer learns the features of both target and background samples to generate discriminative information. The output value after passing through the sigmoid function falls within the range of [0, 1], enabling it to be interpreted as the probability assigned by the detection network to determine whether a pixel represents the target.

To adaptively extract feature representations of a specific TD dataset, the parameters of the detection network should be fine-tuned according to the selected samples from the TD dataset. The fine-tuning strategy consists of three components: freezing a part of layer parameters, retraining a part of layer parameters with random initialization, and continuing to train a part of layer parameters with the pretrained initialization. First, the images of the pretraining task and the downstream task may not be collected by the same sensor, leading to a mismatch in spectral dimensions. The layers related to the number of bands in each path for the DPN encoder (conv2 and conv3) are, therefore, retrained with random initialization. As the detector is primarily used for downstream tasks, its parameters also need to be randomly initialized. Then, DPN encoder has a hierarchical structure [61], where low-level features, including edges and shapes obtained from shallow layers are universal [62]. The front few convolutional layers (conv1, the first three convolutional layers of 1-D block and 2-D block) can, therefore, be frozen and directly transferred to downstream tasks. Because of the limited number of pretraining HSIs, the pretraining model cannot extract all ground features, and the extracted deep features are usually incomplete [63]. The last few layers (the last convolutional layer of 1-D block and 2-D block, GAP layer, and BN layer) should, therefore, continue to update the parameters based on downstream image scenes; in other words, the corresponding parameters in the pretrained DPN encoder can serve as the initialization.

3) BCE Loss: Since the TD task is essentially a binary classification task, the most common binary cross entropy (BCE) loss is used to fine-tune the detection network. The loss function is denoted as follows:

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}[l_i \times \log(p_i) + (1 - l_i) \times \log(1 - p_i) \tag{9}$$

where l_i is the true label of the training sample, and p_i is the output value of the training sample via the detector. L_{BCE} is used to measure the difference between l_i and p_i, which should be reduced during the fine-tuning process.

IV. Experiment

A. Experiment Datasets

Our work uses one hyperspectral dataset for pretraining and four hyperspectral datasets for downstream TD tasks.

1) San Diego Dataset [11]: The San Diego dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with a size of 400 × 400 pixels, and the image is located at San Diego airport in California, United States, as shown in Fig. 4(a). The AVIRIS sensor collects spectral data from 224 bands ranging from 400 to 2500 nm. After removing the water vapor absorption and noise bands, there are 189 remaining bands. The spatial resolution is 3.5 m. Because of the wide variety of ground objects in this image, such as roofs, bare soil, grasslands, roads, airport runways, shadows, etc., this dataset is used for pretraining.

2) AVIRIS Dataset: Because of the difficulty in obtaining ground truth for each aircraft in this image, some scholars [30], [32], [37], [39] used subimage, including three aircraft as a dataset for the TD task. We crop 100 × 100 pixels in the middle of the image as the first downstream task dataset, as shown in Fig. 4(b). Fig. 4(c) illustrates the ground-truth image of three aircrafts, consisting 134 target pixels. The ratio of the number of target pixels of interest to the total number of pixels in the image is 1.34%. There are three prior target spectra.

3) Muufl Dataset [64]: The image is located in Long Beach, MS, United States, and is part of the Gulf Park campus of the University of Southern Mississippi. In this image, the wavelength range is 367.7–1043.4 nm, with 72 spectral bands. There are 64 remaining bands after discarding the noise bands. The image size is 325 × 220 pixels, as shown in Fig. 4(d), and Fig. 4(e) illustrates the ground-truth image of several cloth boards consisting of 269 pixels. The ratio of the number of target pixels of interest to the total number of pixels in the image is 0.414%. There are two prior target spectra.

4) Nuance Cri Dataset [65]: The Nuance Cri dataset was collected by the Nuance Cri hyperspectral sensor with a size of 400 × 400 pixels, as shown in Fig. 4(f). The wavelength range of this data set is 650–1100 nm, with 46 spectral bands. Fig. 4(g) illustrates the ground-truth image of ten rock targets, consisting of 1254 pixels. The ratio of the number of interested target pixels to the total number of pixels in the image is 0.783%. There are two prior target spectra.

5) HYDICE Dataset [66]: The HYDICE dataset was collected by the hyperspectral digital imagery collection experiment (HYDICE) sensor with a size of 150 × 85 pixels,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

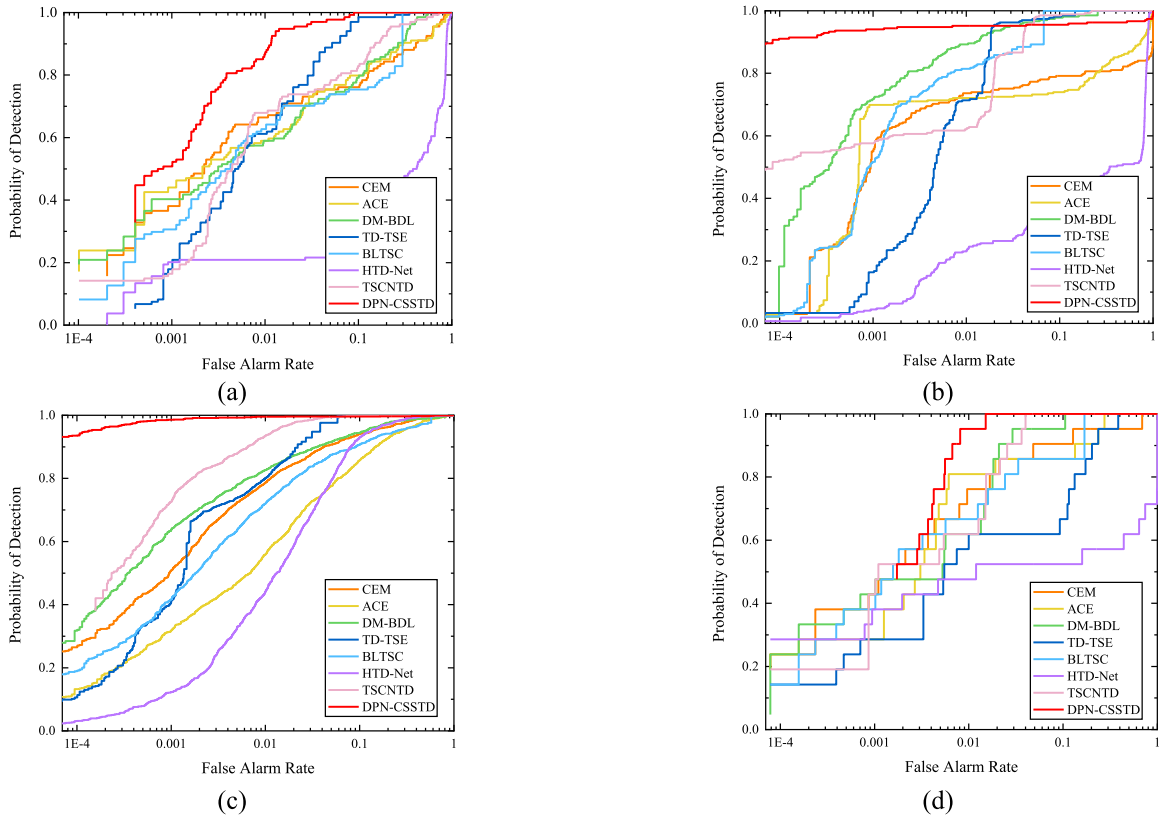


Fig. 5. ROC curves of different methods for four datasets. (a) AVIRIS. (b) Muufl. (c) Nuance Cri. (d) HYDICE.

as shown in Fig. 4(h). The image is located in a suburban residential area of Fort Hood, TX, United States. The dataset has 210 bands, but only 162 bands remain after removing noise bands. The spectral resolution is 10 nm, and the spatial resolution is 2 m. Fig. 4(i) illustrates the ground-truth image of the vehicle targets, consisting of 21 pixels. The ratio of the number of interested target pixels to the total number of pixels in the image is 0.165%. There are three prior target spectra.

B. Experimental Settings

In order to verify the detection performance of the proposed method, we selected six detection methods for comparison, including two classic detectors, CEM [9] and ACE [13], two machine learning methods DM-BDL [21] and TD-TSE [23], and three DL methods BLTSC [35], HTD-Net [29], and TSCNTD [30]. Through experiments and referring to the original texts, all comparison methods used the optimal parameters. To alleviate the influence of random factors, we conducted five consecutive runs for both the proposed method and the DL-based comparison methods. The value of each evaluation criterion ranked in third is taken as the final evaluation value.

For the DPN-CSSTD method, the learning rate, the epoch, and the batch size of the pretrained network are set as 1e-3, 60, and 256, respectively. The learning rate, the epoch, and the batch size of the fine-tuning network are set as 1e-3, 200, and the total number of selected samples. The optimizer used is Adam. All the experiments are performed in the 64-bit Windows 10 operating system (OS) under the NVIDIA GeForce RTX 3060 graphics processing unit (GPU), and 16-GB random access memory.

To evaluate the detection performance, we used four quantitative criteria [67], [68], namely the receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC), the separability map, and the running time.

C. Detection Performance

This section analyzes the detection performance of the proposed DPN-CSSTD method compared with other detection methods. We evaluate the effectiveness of transferring the pretrained model to various TD tasks with four different datasets. DM-BDL is a typical sparse representation-based method, which is the abbreviation of a decomposition model with background dictionary learning for hyperspectral TD (HTD) [21]. TD-TSE is the abbreviation of the target detector through tree-structured encoding, which is a typical tree-structure-based method [23]. The experimental results are provided via four criteria.

The ROC curves of different methods for four datasets are shown in Fig. 5. It is evident that the DPN-CSSTD method achieves the optimal ROC curve on most of the datasets, where it is closest to the upper left corner. For the Muufl dataset, when the false alarm rate is less than 0.01, the ROC curve of DPN-CSSTD is higher than the other curves. In addition, it can be seen that the ROC curves of the other DL-based detectors on four datasets are relatively low.

The AUC values of different methods for four datasets are shown in Table I, where the maximum AUC values and the minimum deviations are in bold. DPN-CSSTD shows the maximum AUC values on all four datasets. Both DPN-CSSTD and BLTSC demonstrate a maximum AUC deviation on two datasets; however, the AUC values of DPN-CSSTD are better than that of BLTSC. The results, therefore,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I
AUC VALUES AND DEVIATIONS OF DIFFERENT METHODS FOR FOUR DATASETS

	CEM	ACE	DMBDL	TDTSE	BLTSC	HTD-Net	TSCNTD	DPN-CSSTD
AVIRIS	0.9004	0.9041	0.9350	0.9787	0.8680±0.0029	0.5533±0.0825	0.9479±0.2420	**0.9950±0.0025**
Muufl	0.8187	0.8520	0.9864	0.9832	0.9004±**0.0016**	0.5476±0.3000	0.9874±0.0050	**0.9946**±0.0069
Nuance Cri	0.9767	0.9481	0.9787	0.9818	0.9573±0.0023	0.9647±0.1205	0.9970±0.0021	**0.9984±0.0011**
HYDICE	0.9552	0.9663	0.9878	0.9271	0.9114±**0.0015**	0.6194±0.1279	0.9904±0.0038	**0.9966**±0.0035

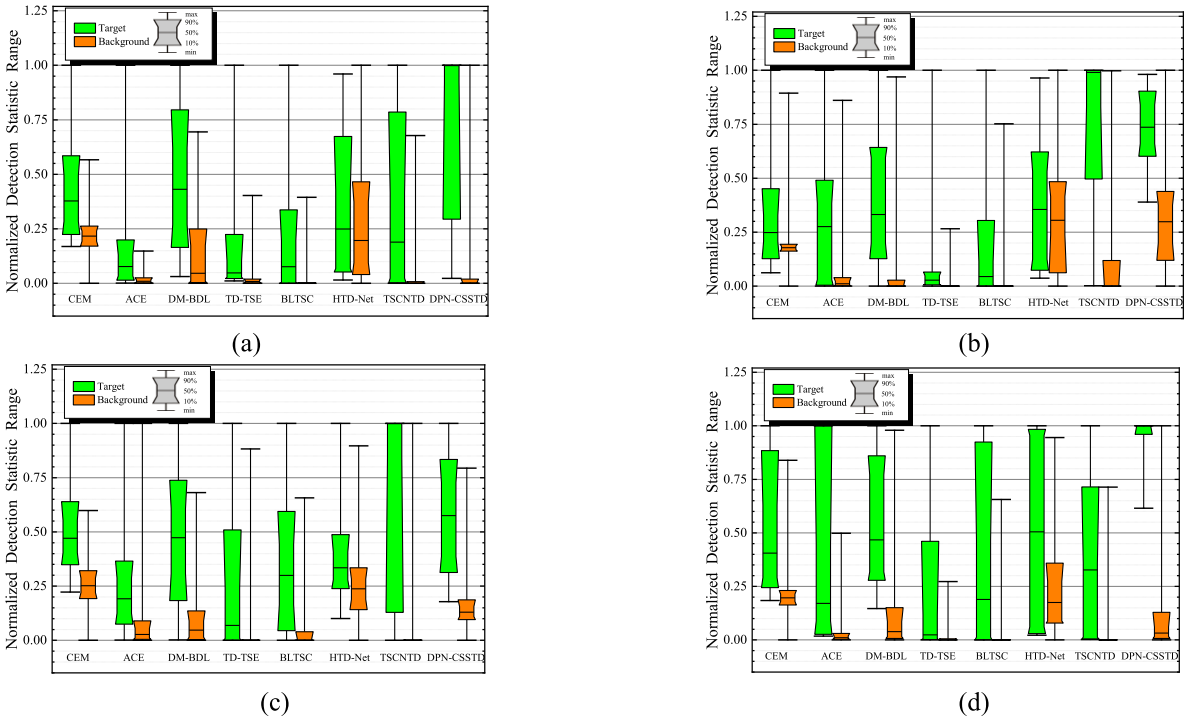


Fig. 6. Separability maps of different methods on four datasets. (a) AVIRIS. (b) Muufl. (c) Nuance Cri. (d) HYDICE.

indicate that DPN-CSSTD has a robust and superior detection performance.

The separability maps of each method for four datasets are shown in Fig. 6, which are to verify the effectiveness of the proposed method in separating the target from the background. From the results, it can be clearly seen that the gap between the green boxes (target) and orange boxes (background) corresponding to the proposed method is larger than that of other methods and has superior separability. In addition, the green box of our method is generally in a higher position, indicating that DPN-CSSTD can better highlight the targets.

The runtime of machine learning and DL methods for four datasets is shown in Table II. Among them, for the DL-based detection methods, the training time and testing time are provided, respectively, whereas our method provides the training time for fine-tuning. From the results in the table, it can be seen that machine learning methods take less time than other DL methods except the Nuance Cri dataset. It is worth noting that DPN-CSSTD used less time than machine learning methods except for the AVIRIS dataset. Among four DL methods, the proposed DPN-CSSTD method greatly saves time, especially the training time, which is due to the use of the transfer learning model that greatly reduces the number of parameters to be trained.

The results of different evaluation criteria, therefore, show that the proposed DPN-CSSTD method has a robust and superior detection performance compared with other detection methods on four datasets. These results also demonstrate that our CSSL pretrained model really learns image features of specific HSI that are effectively transferred to alternative TD tasks, saving the time consumption of training multiple TD datasets.

D. Parameter Analysis and Ablation Analysis

1) Parameter Analysis: This section analyzes two key parameters of the proposed DPN-CSSTD method, namely the number of background samples and the number of target samples. First, the numbers of target and background samples are changed together to determine a rough parameter value for the single variable method. Considering the minimum number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RUNNING TIME (IN SECONDS) OF MACHINE AND DEEP LEARNING METHODS FOR FOUR DATASETS

	DM-BDL	TD-TSE	BLTSC		HTD-Net		TSCNTD		DPN-CSSTD	
			Train	Test	Train	Test	Train	Test	Train	Test
AVIRIS	3.485	2.635	598.263	0.487	1159.574	2.890	170.306	3.189	3.890	1.526
Muufl	8.365	36.428	1948.148	2.382	447.941	2.896	447.941	3.956	1.721	4.475
Nuance Cri	54.933	110.888	3504.171	6.163	103.253	2.928	26.427	5.661	1.831	8.351
HYDICE	4.791	3.687	475.851	0.533	1136.385	2.976	145.546	3.581	1.956	1.575

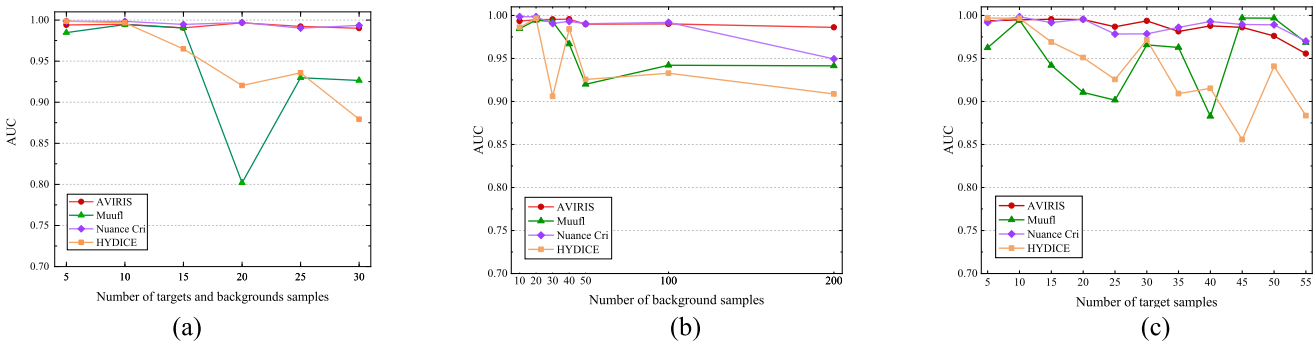


Fig. 7. Parameter analysis on four datasets. (a) Number of target and background samples. (b) Number of background samples. (c) Number of target samples.

of the target in all datasets is 21 and the proposed method is to use a small number of samples to fine-tune the transfer learning model, we set the number of samples within 30. The rough range of two parameters is set as [5, 10, 15, 20, 25, 30].

In order to alleviate the influence of random factors, the AUC value ranked third from five consecutive runs is visualized, as shown in Fig. 7(a). On the four datasets, the AUC curves show a general trend of first increasing and then decreasing, especially for the Muffl and HYDICE datasets, and when the parameter value is 10, DPN-CSSTD performs well on all datasets.

We, therefore, fixed the number of target samples as 10 and further analyzed the influence of the number of background samples on the final detection performance. The rough range of the number of background samples is set as [10, 20, 30, 40, 50, 100, 200]. Similarly, the median AUC value obtained from five consecutive runs is visualized, as shown in Fig. 7(b). On the four datasets, the AUC curves show an overall trend of first slowly increasing and then decreasing, and when the number of background samples is 20, DPN-CSSTD performs well on all four datasets.

Then, we conduct reverse validation by setting the number of background samples as 20 and the rough range of the number of target samples as [5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55]. Similarly, the AUC curves in Fig. 7(c) show a general trend of first increasing and then decreasing on the four datasets, and when the number of target samples is 10, a DPN-CSSTD performs well on all four datasets.

The above results, therefore, recommend a universal setting of 10 and 20 for the number of target and background samples on all four datasets, respectively.

2) Ablation Analysis: This section analyzes the effectiveness of the spatial-spectral contrastive self-supervised pretraining, the SLIC segmentation strategy, and the dual path networks.

First, we analyze the effectiveness of the spatial-spectral contrastive self-supervised pretraining on four TD datasets. The experiments aim to state the effectiveness of image representation learned by the spatial-spectral contrastive self-supervised pretraining model with a DPN encoder when transferred to different downstream tasks. We compare the pretrained DPN with the randomly initialized DPN when transferred to a TD task with the same training samples and fine-tuning strategy, labeled as DPN-CSSTD and DPN-RITD. In order to alleviate the influence of random factors, the median AUC value obtained from five consecutive runs is visualized, as shown in Fig. 8(a). With the contrastive self-supervised pretraining, the AUC value improves on the four datasets and specially increases by 0.0171 and 0.0313 in the Muufl dataset and the HYDICE dataset, respectively.

Then, we evaluate the effectiveness of the SLIC segmentation strategy on four TD datasets. The experiments aim to statement the effectiveness of background samples selected by the SLIC segmentation strategy when fine-tuning the transfer learning module. We compare the DPN-CSSTD via SLIC segmentation strategy, random selection strategy, and initial detection strategy. The random selection strategy is to randomly select 20 background samples from the entire image, and the initial detection strategy is to select 20 pixels with smallest detection results via CEM. The median AUC value obtained from five consecutive runs is visualized, as shown in Fig. 8(b). On the four datasets, DPN-CSSTD via random selection strategy outperforms DPN-CSSTD via initial detection strategy, and DPN-CSSTD via SLIC segmentation strategy outperforms that via random selection strategy. The reason may lie in the representativeness of background samples selected through three strategies. The background samples selected through the initial detection strategy may not cover various background objects. The SLIC segmentation strategy can select one representative background sample from each cluster that differs from the a priori target spectra, exhibiting more stability than the random selection strategy.

Finally, we analyze the effectiveness of the dual path networks DPN on four TD datasets. We compare DPN with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

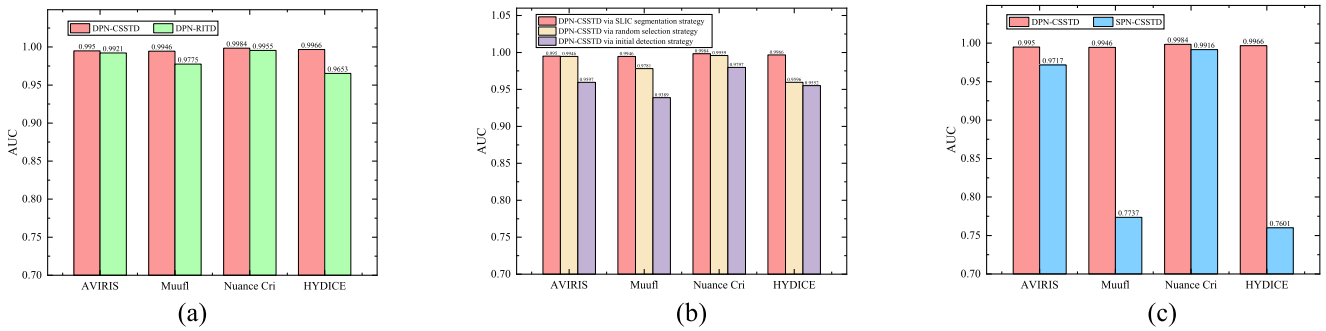


Fig. 8. Ablation analysis on four datasets. (a) The spatial-spectral contrastive self-supervised pretraining.(b) The SLIC segmentation strategy. (c) The dual path networks.

the single spatial path network (SPN) in feature extraction. The feature representations extracted via DPN and SPN are transferred to the downstream TD task, labeled as DPN-CSSTD and SPN-CSSTD. The median AUC value obtained from five consecutive runs is visualized, as shown in Fig. 8(c). On four datasets, the AUC values of DPN-CSSTD increased by 0.0233, 0.2209, 0.0068, and 0.2365 compared to SPN-CSSTD, respectively. This experimental result demonstrates that the combination of the spectral path network and the SPN indeed helps to improve the feature extraction, which significantly contributes to the improvement in detection performance.

V. Conclusion

This article proposes a DPN-CSSTD method for HTD, which exhibits superior performance on four datasets. By constructing a spatial-spectral contrastive self-supervised pretraining model based on an HSI augmentation method and a DPN encoder, the pretraining model can be trained with spatial-spectral discrimination ability without labeled data, and by selecting a few representative target and background samples, the transfer learning model with the pretrained dual path networks and a newly added detector can be effectively applied to various TD tasks through a fine-tuning strategy.

The proposed method, however, still has deficiencies in background suppression and edge preservation. The pretraining dataset may not be large enough, resulting in the fact that the background suppression effect is not so good on specific dataset. In addition, the SPN implements convolution operation on regular image patches, which cannot preserve the boundary features of the targets and the backgrounds on some datasets, resulting in some false alarms and missed targets. We will study how to better learn the image features and adapt the pretrained model to numerous TD tasks.

Acknowledgment

The authors would like to thank all the editors and anonymous reviewers for their careful reading and insightful remarks.

References

[1] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[2] C. Jiao et al., "Sparsity-regularized attention multiple-instance network for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 124–137, Jan. 2023.

[3] C. Ye et al., "Landslide detection of hyperspectral remote sensing data based on deep learning with constrains," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 5047–5060, Dec. 2019.

[4] R. A. Witkosky et al., "The lavic lake fault: A long-term cumulative slip analysis via combined field work and thermal infrared hyperspectral airborne remote sensing," *Remote Sens.*, vol. 12, no. 21, p. 3586, Nov. 2020.

[5] C. Ke, "Military object detection using multiple information extracted from hyperspectral imagery," in *Proc. Int. Conf. Prog. Informat. Comput.*, 2017, pp. 124–128.

[6] K. Sendin, P. J. Williams, and M. Manley, "Near infrared hyperspectral imaging in quality and safety evaluation of cereals," *Crit. Rev. Food Sci. Nutrition*, vol. 58, no. 4, pp. 575–590, Mar. 2018.

[7] X. Hadoux et al., "Non-invasive in vivo hyperspectral imaging of the retina for potential biomarker use in Alzheimer's disease," *Nature Commun.*, vol. 10, no. 1, p. 4227, Sep. 2019.

[8] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[9] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Univ. Maryland, Baltimore, MD, USA, 1993.

[10] J. C. Harsanyi and C.-I. Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.

[11] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.

[12] R. T. Behrens and L. L. Scharf, "Signal processing applications of oblique projection operators," *IEEE Trans. Signal Process.*, vol. 42, no. 6, pp. 1413–1424, Jun. 1994.

[13] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.

[14] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, May 1992.

[15] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Trans. Signal Process.*, vol. 42, no. 8, pp. 2146–2157, Aug. 1994.

[16] H. Kwon and N. M. Nasrabadi, "Kernel spectral matched filter for hyperspectral imagery," *Int. J. Comput. Vis.*, vol. 71, no. 2, pp. 127–141, Feb. 2007.

[17] Y. Zhang, L. Zhang, B. Du, and S. Wang, "A nonlinear sparse representation-based binary hypothesis model for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2513–2522, Jun. 2015.

[18] Y. Dong, B. Du, and L. Zhang, "Target detection based on random forest metric learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 4, pp. 1830–1838, Apr. 2015.

[19] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2014.

[20] Y. Zhang, B. Du, and L. Zhang, "A sparse representation-based binary hypothesis model for target detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1346–1354, Mar. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

[21] T. Cheng and B. Wang, "Decomposition model with background dictionary learning for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1872–1884, 2021.

[22] D. Zhu, B. Du, and L. Zhang, "Learning single spectral abundance for hyperspectral subpixel target detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jan. 30, 2023, doi: 10.1109/TNNLS.2023.3239061.

[23] X. Sun et al., "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, May 2021.

[24] L. Zhang, L. Zhang, D. Tao, and X. Huang, "Sparse transfer manifold embedding for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 1030–1043, Feb. 2014.

[25] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[26] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

[27] D. Wang, J. Zhang, B. Du, L. Zhang, and D. Tao, "DCN-T: Dual context network with transformer for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 2536–2551, 2023.

[28] S. Feng, R. Feng, J. Liu, C. Zhao, F. Xiong, and L. Zhang, "An attention-based multiscale spectral–spatial network for hyperspectral target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 5503705.

[29] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[30] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[31] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[32] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022.

[33] H. Gao, Y. Zhang, Z. Chen, S. Xu, D. Hong, and B. Zhang, "A multi-depth and multibranch network for hyperspectral target detection based on band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506818.

[34] Y. Li, H. Qin, and W. Xie, "HTDFormer: Hyperspectral target detection based on transformer with distributed learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524715.

[35] W. Xie, X. Zhang, Y. Li, K. Wang, and Q. Du, "Background learning based on target suppression constraint for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5887–5897, 2020.

[36] Y. Gao, Y. Feng, and X. Yu, "Hyperspectral target detection with an auxiliary generative adversarial network," *Remote Sens.*, vol. 13, no. 21, p. 4454, Nov. 2021.

[37] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[38] X. Zhang et al., "Self-supervised learning with deep clustering for target detection in hyperspectral images with insufficient spectral variation prior," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103405.

[39] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[40] Y. Shi, J. Li, Y. Li, and Q. Du, "Sensor-independent hyperspectral target detection with semisupervised domain adaptive few-shot learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6894–6906, Aug. 2021.

[41] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527913.

[42] I. Misra and L. van der Maaten, "Self-supervised learning of pretext-invariant representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6706–6716.

[43] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[44] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," 2019, *arXiv:1911.05722*.

[45] J.-B. Grill et al., "Bootstrap your own latent: A new approach to self-supervised learning," 2020, *arXiv:2006.07733*.

[46] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," 2020, *arXiv:2002.05709*.

[47] S. Lee, D. Bok Lee, and S. Ju Hwang, "Contrastive learning with adversarial perturbations for conditional text generation," 2020, *arXiv:2012.07280*.

[48] Y. Qu, D. Shen, Y. Shen, S. Sajeev, J. Han, and W. Chen, "CoDA: Contrast-enhanced and diversity-promoting data augmentation for natural language understanding," 2020, *arXiv:2010.08670*.

[49] O. J. Hénaff et al., "Data-efficient image recognition with contrastive predictive coding," 2019, *arXiv:1905.09272*.

[50] Y. Wang, C. M. Albrecht, N. A. A. Braham, L. Mou, and X. X. Zhu, "Self-supervised learning in remote sensing: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 10, no. 4, pp. 213–247, Dec. 2022.

[51] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6008205.

[52] X. Kang, B. Deng, P. Duan, X. Wei, and S. Li, "Self-supervised spectral–spatial transformer network for hyperspectral oil spill mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507410.

[53] X. Ou, L. Liu, S. Tan, G. Zhang, W. Li, and B. Tu, "A hyperspectral image change detection framework with self-supervised contrastive learning pretrained model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7724–7740, 2022.

[54] Y. Can, Y. Yuan, and J. Zhiyu, "Self-supervised spectral matching network for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Oct. 2021, pp. 2524–2527.

[55] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[56] A. Karpathy, G. Toderici, S. Shetty, T. Leung, R. Sukthankar, and L. Fei-Fei, "Large-scale video classification with convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1725–1732.

[57] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[58] Y. LeCun et al., "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.

[59] X. Ren and J. Malik, "Learning a classification model for segmentation," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, vol. 1, Oct. 2003, pp. 10–17.

[60] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[61] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 7553.

[62] J. Yang, Y.-Q. Zhao, and J. C. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[63] M. Oquab, L. Bottou, I. Laptev, and J. Sivic, "Learning and transferring mid-level image representations using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1717–1724.

[64] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep., REP-2013-570, 2013.

[65] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[66] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.

[67] X. Sun, N. Li, and H. Zhao, "Performance evaluation for hyperspectral target detection algorithms," *Proc. SPIE*, vol. 7127, pp. 712725–712726, Oct. 2008.
[68] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.



Xi Chen received the B.E. degree in Earth information science and technology from the School of Geophysics and Geomatics, China University of Geosciences, Wuhan, China, in 2022, where she is currently pursuing the M.S. degree in resources and environment with the School of Geophysics and Geomatics.

Her research interests include hyperpectral target detection and transfer learning.



Yuxiang Zhang (Senior Member, IEEE) received the B.S. degree in the sciences and techniques of remote sensing and the Ph.D. degree in the photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

She was a Visiting Scholar with The University of Sydney, Sydney, NSW, Australia, from 2019 to 2020. She is currently an Associate Professor with the School of Geophysics and Geomatics, China University of Geosciences, Wuhan. Her research interests include hyperspectral image processing, machine learning, and deep learning.

Dr. Zhang regularly serves as a PC Member of IJCAI and AAAI. She serves as a reviewer of more than 20 international journals, including the IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and PR.



Yanni Dong (Senior Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing from Wuhan University, Wuhan, China, in 2012, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote sensing, Wuhan University, in 2017.

She is currently a Professor with the School of Resource and Environmental Sciences, Wuhan University. She was a Hong Kong Scholar with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong. Her research interests include hyperspectral image processing, pattern recognition, and machine learning.

Dr. Dong regularly serves as PC Member of IJCAI and AAAI. She serves as a reviewer of more than 20 international journals, including the IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and PR. She also serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Bo Du (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2010.

He is currently a Professor with the School of Computer Science and the Institute of Artificial Intelligence, Wuhan University. He is also the Director of the National Engineering Research Center for Multimedia Software, Wuhan University. He has more than 80 research papers published in the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS). Fourteen of them are ESI hot papers or highly cited papers. His major research interests include machine learning, computer vision, and image processing.

Dr. Du regularly serves as a Senior PC Member of IJCAI and AAAI. He won the Highly Cited Researcher (2019\2020) by the Web of Science Group. He won the IEEE Geoscience and Remote Sensing Society 2020 Transactions Prize Paper Award. He won the International Joint Conferences on Artificial Intelligence (IJCAI) Distinguished Paper Prize, IEEE Data Fusion Contest Champion, and IEEE Workshop on Hyperspectral Image and Signal Processing Best Paper Award, in 2018. He served as the Area Chair for ICPR. He serves as an Associate Editor for *Neural Networks*, *Pattern Recognition*, and *Neurocomputing*. He also serves as a reviewer for 20 Science Citation Index (SCI) magazines, including IEEE TPAMI, IEEE TCYB, IEEE TGRS, IEEE TIP, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:51:45 UTC from IEEE Xplore. Restrictions apply.